

TAMIL NADU STATE BOARD - STANDARD 11 MATHEMATICS VOLUME 2

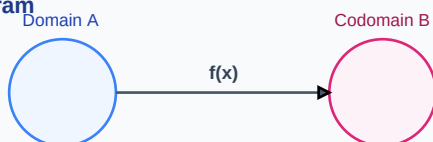
CHAPTER 8: VECTOR ALGEBRA

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 11 English Medium
Subject:	Mathematics Volume 2
Chapter Name:	Vector Algebra

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Function Mapping Diagram

Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Unit Vector	A vector with a magnitude of exactly one, pointing in a specific directional orientation, denoted by $\hat{a}$.
Scalar Product	The product of the magnitudes of two vectors and the cosine of the angle between them, yielding a scalar.
Direction Cosines	The cosines of the angles made by a vector with the three positive coordinate axes (x, y, z).

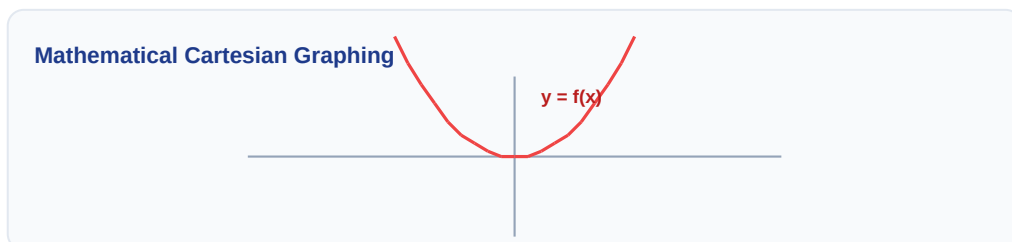
Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

Venn Diagram: $A \cap B$



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:



Essential Formulas, Laws & Identities:

- Dot Product: $a \cdot b = |a| \cdot |b| \cdot \cos(\theta)$
- Cross Product: $a \times b = |a| \cdot |b| \cdot \sin(\theta) \cdot \hat{n}$
- Section Formula: $r = \frac{m \cdot b + n \cdot a}{m + n}$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Find the angle between vectors $\mathbf{a} = 2\mathbf{i} + \mathbf{j} - \mathbf{k}$ and $\mathbf{b} = \mathbf{i} + \mathbf{j}$.

Step-by-step Solution:

$$\mathbf{a} \cdot \mathbf{b} = 2(1) + 1(1) + (-1)(0) = 3.$$

$$|\mathbf{a}| = \sqrt{4+1+1} = \sqrt{6} \quad |\mathbf{b}| = \sqrt{1+1} = \sqrt{2}.$$

$$\cos(\theta) = \mathbf{a} \cdot \mathbf{b} / (|\mathbf{a}||\mathbf{b}|) = 3 / (\sqrt{6} \cdot \sqrt{2}) = 3 / \sqrt{12} = \sqrt{3}/2 \Rightarrow \theta = 30 \text{ degrees}.$$

Linear Algebraic Matrix Representation

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Prove by vector method that the diagonals of a rhombus bisect each other at right angles.

Analytical Solution Strategy:

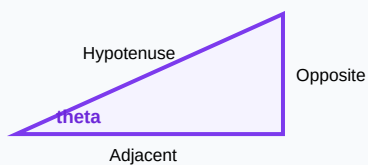
Let O be origin. Vector diagonals are $d_1 = a + b$ and $d_2 = a - b$.

rhombus condition: $|a| = |b|$.

$d_1 \cdot d_2 = (a + b) \cdot (a - b) = |a|^2 - |b|^2 = 0$.

Since the dot product is zero, the diagonals are perpendicular.

Right-Angled Triangle Geometry



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Find the unit vector along the vector $a = 3i - 4j + 12k$.

Q2 (2-Mark): Find the projection of vector $i - j$ on $i + j$.

Q3 (2-Mark): Calculate the work done by force $F = 2i + 3j - k$ in displacing a body from $(1, 1)$ to $(3, 4)$.

Q4 (5-Mark): Determine the value of λ for which vectors $a = 2i + \lambda j + k$ and $b = i - 2j + 3k$ are perpendicular.

Q5 (5-Mark): Calculate the torque vector for force $F = 3i + j - 2k$ acting at point $r = i + j$.

Circle with Radius Vector on Plane



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Flight Dynamics

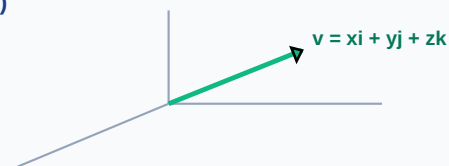
Airplanes use vector calculations to determine crosswind drift angles and adjust compass headings.

Application Case: Game Physics

Vectors calculate physics paths, light reflections, and bullet trajectories in game engines.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

3D Vector Projections (x, y, z)



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Probability Tree Outcome Diagram

